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Bubble functions prompt unusual stabilized finite element methods

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Abstract

A second-order linear scalar differential equation including a zero-th order term is approximated using first the standard Galerkin method enriched with bubble functions. Static condensation of the bubbles suggest an unusual stabilized finite element method for which we establish a convergence study and obtain successful numerical simulations. The method is generalized to allow for a convection operator in the equation. This work may be employed as a starting point for simulation of non-linear equations governing turbulence phenomena, flows with chemical reactions, and other important problems.

1. Introduction

We have pointed out in a recent communication [9] that for a certain model problem, bubble functions added to the usual finite element polynomials seem to subtract stability from the formulation. This finding contrasts our experience with other models such as the advective–diffusive equations and saddle-point problems [2, 5, 6, 20, 21], where bubble functions are key to obtaining stable formulations.

In this note we show that the apparent contradiction raised in [9] is in fact an inspiration to develop *unusual* stabilized methods. The subtraction prescribed by the static condensation of the bubbles indicates that stability can be achieved in a non-standard manner by stabilized methods that we present herein.

In Section 2, we describe the model problem, the approximation of its solution with a Galerkin method using piecewise linear polynomials enriched with bubble functions, and the consequences of eliminating the bubbles. In Section 3 we present a stabilized method and its error analysis. In Section 4 we add convection to the model equation and suggest a stabilized method to deal with all effects simultaneously. Finally, we report on some numerical simulations that confirm the arguments presented herein.

2. A model problem and bubbles' effects

We start with the model problem given by: find a scalar valued function $u(x)$ defined in $\Omega \subset \mathbb{R}^2$ such that

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$$\sigma u - \kappa \Delta u = f \quad \text{in } \Omega \quad (1)$$

$$u = 0 \quad \text{on } \Gamma = \partial\Omega \quad (2)$$

where σ and κ are given positive constants and $f(x)$ is a given source function assumed to be square-integrable in Ω (i.e. $f \in L_2(\Omega)$).

The variational formulation corresponding to (1–2) is: find $u \in H_0^1(\Omega)$ such that

$$(\sigma u, v) + (\kappa \nabla u, \nabla v) = (f, v) \quad v \in H_0^1(\Omega) \quad (3)$$

where $H_0^1(\Omega)$ is the Hilbert space of functions with square-integrable value and derivative in Ω satisfying (2), and we use the notation $(f, g) = \int_{\Omega} fg \, d\Omega$ or $(f, g) = \int_{\Omega} f \cdot g \, d\Omega$ depending on whether the underlying inner product is performed between scalar or vector fields, respectively.

The standard Galerkin method is formulated on a subspace $V_h \subset H_0^1(\Omega)$ employing the variational formulation (3): find $u_h \in V_h$ such that

$$(\sigma u_h, v) + (\kappa \nabla u_h, \nabla v) = (f, v) \quad v \in V_h \quad (4)$$

We consider the following subspace $V_h = V_h^b$

$$V_h^b = \{v \in H_0^1(\Omega) \mid v|_K \in P_1(K) \oplus B(K), K \in \mathcal{C}_h\}, \quad (5)$$

where $\mathcal{C}_h$ is a partition of Ω into regularly shaped triangles, $P_1(K)$ denotes the space of linear functions defined on the triangle K and $B(K)$ denotes the space of bubble functions (e.g. spanned by a cubic function). The bubble basis function $\varphi \in B(K)$ satisfies

$$\begin{aligned} \varphi(x) &> 0 \quad \forall x \in K \\ \varphi(x) &= 0 \quad \forall x \in \partial K \end{aligned} \quad (6)$$

and $\varphi = 1$ at the baricenter of the triangle.

We now proceed to eliminate the bubbles from (4) using the static condensation procedure. First we obtain the ‘bubble equation’ by selecting in (4) $v = \varphi(x)$ for $x \in K$ and $v = 0$ elsewhere in Ω to obtain

$$(\sigma u_h, \varphi)_K + (\kappa \nabla u_h, \nabla \varphi)_K = (f, \varphi)_K \quad (7)$$

where the subscript K indicates integration over K .

Denoting by V_1 the space of piecewise linears, i.e.

$$V_1 = \{v \in H_0^1(\Omega) \mid v|_K \in P_1(K), K \in \mathcal{C}_h\} \quad (8)$$

we may now decompose the unknown solution to (4) u_h into its linear part $u_1 \in V_1$ and its part spanned by the bubble, i.e.

$$u_h = u_1 + \sum_{K \in \mathcal{C}_h} u_b^K \varphi \quad (9)$$

where u_b^K is the unknown bubble coefficient.

Substituting (9) into (7) we get

$$(\sigma u_1, \varphi)_K + u_b^K (\sigma \varphi, \varphi)_K + (\kappa \nabla u_1, \nabla \varphi)_K + u_b^K (\kappa \nabla \varphi, \nabla \varphi)_K = (f, \varphi)_K. \quad (10)$$

However, $\forall w_1 \in V_1$ we have by integration by parts

$$(\nabla w_1, \nabla \varphi)_K = -(\Delta w_1, \varphi)_K + (\nabla w_1 \cdot \mathbf{n}, \varphi)_{\partial K} = 0 \quad w_1 \in V_1 \quad (11)$$

Therefore, using (11) for $w_1 = u_1$, we reduce (10) to

$$u_b^K [\sigma \|\varphi\|_{0,K}^2 + \kappa \|\nabla \varphi\|_{0,K}^2] = (f - \sigma u_1, \varphi)_K, \quad (12)$$

where we have used the notation $\|v\|_{0,K}^2 = \int_K v^2 \, d\Omega$. Solving (12) for the bubble coefficient in each element leads to

$$u_b^K = \frac{-1}{\sigma \|\varphi\|_{0,K}^2 + \kappa \|\nabla \varphi\|_{0,K}^2} (\sigma u_1 - f, \varphi)_K. \quad (13)$$

The procedure can be repeated to each element $K \in \mathcal{C}_h$ and Eq. (13) gives the value of each unknown coefficient u_b^K as a function of the chosen bubble basis function φ , the data $(\sigma, \kappa$ and $f)$ and the unknown linear part u_1 of the solution u_h . The next step of static condensation consists in selecting in (4) $v = v_1 \in V_1$, and by using (9) and (11) we get

$$(\sigma u_1, v_1) + \sum_{K \in \mathcal{C}_h} u_b^K (\sigma \varphi, v_1)_K + (\kappa \nabla u_1, \nabla v_1) = (f, v_1) \quad v_1 \in V_1 \quad (14)$$

Therefore, the resulting variational equation (14) is equivalent to using the standard Galerkin method for piecewise linear functions (i.e. choosing $V_h = V_1$ in (4)) ‘plus’ a term that in view of (13) can be written as

$$\sum_{K \in \mathcal{C}_h} u_b^K (\sigma \varphi, v_1)_K = - \sum_{K \in \mathcal{C}_h} \frac{1}{\sigma \|\varphi\|_{0,K}^2 + \kappa \|\nabla \varphi\|_{0,K}^2} (\sigma u_1 - f, \varphi)_K (\varphi, \sigma v_1)_K. \quad (15)$$

For the cubic bubble we have (see [20])

$$\int_K \varphi \, d\Omega = C_1 h_K^2, \quad \varphi|_K \in B(K), \quad K \in \mathcal{C}_h \quad (16)$$

$$\|\varphi\|_{0,K} = C_2 h_K^2, \quad \varphi|_K \in B(K), \quad K \in \mathcal{C}_h \quad (17)$$

$$\|\nabla \varphi\|_{0,K} = C_3, \quad \varphi|_K \in B(K), \quad K \in \mathcal{C}_h \quad (18)$$

where C_1 , C_2 and C_3 are positive constants, independent of the element parameter h_K .

Using the results in [4] together with (16)–(18), (15) can be rewritten as

$$\sum_{K \in \mathcal{C}_h} u_b^K (\sigma \varphi, v_1)_K = - \sum_{K \in \mathcal{C}_h} \frac{C h_K^2}{\sigma h_K^2 + \kappa \bar{C}} (\sigma u_1 - f, \sigma v_1)_K. \quad (19)$$

where C and $\bar{C}$ are positive constants.

We now present a numerical simulation of the problem described in Fig. 1, with $\sigma = 1$, $\kappa = 10^{-6}$, $f = 1$, the finite element space described by (5) and bubble functions of the cubic type. After static condensation the resulting elevation plot of the piecewise-linear function is presented in Fig. 2. To contrast, we next take $V_h = V_1$ in (4), and plot the numerical results in Fig. 3. We note that the addition of the bubbles attenuate the oscillations present using the standard Galerkin method with piecewise linears.

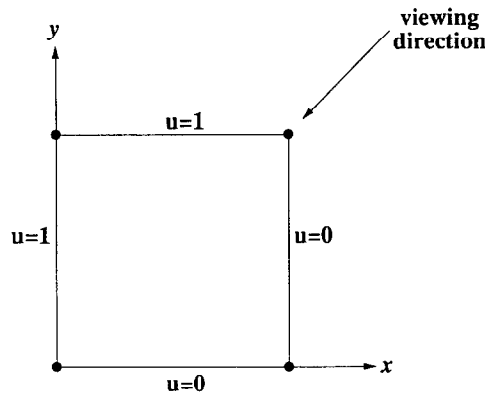


Fig. 1. Boundary conditions for test problem with $f = 1$, $\sigma = 1$ and $\kappa = 10^{-6}$.

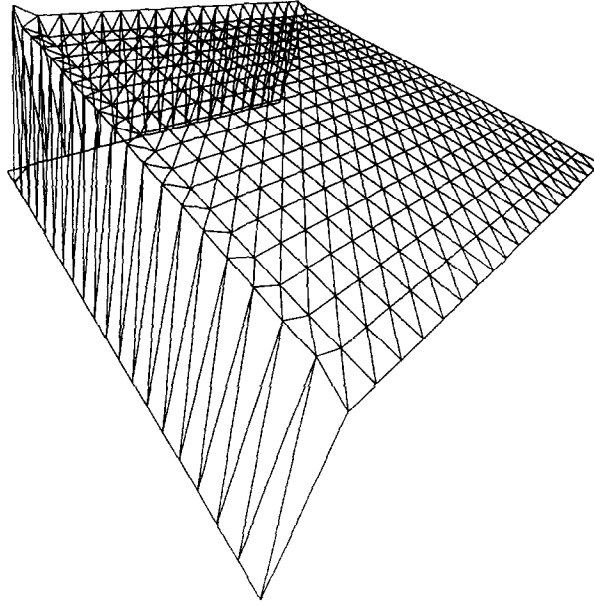


Fig. 2. Elevation plot for the Galerkin method using piecewise linears plus bubbles (the plot shows the results after static condensation).

REMARK 1. The variational formulation (14) governing u_1 , the piecewise-linear part of the solution u_h to (4), indicates that a term should be subtracted from the standard Galerkin method. Furthermore, by substituting v_1 by u_1 we can immediately see that the term

$$\sum_{K \in \mathcal{E}_h} \frac{Ch_K^2}{\sigma h_K^2 + C_K} (\sigma u_1, \sigma u_1)_K$$

is being subtracted from the Galerkin term $(\sigma u_1, u_1)$. In other words, the bubble elimination suggests

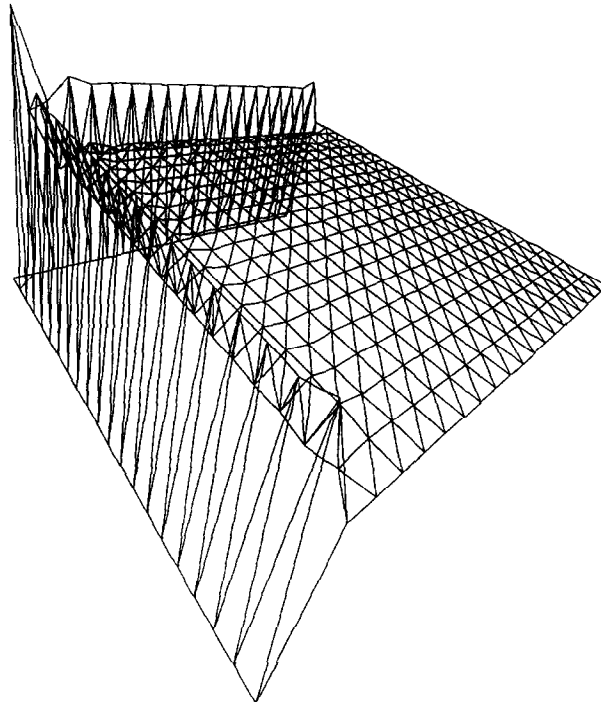


Fig. 3. Elevation plot for the Galerkin method using piecewise linears.